Assignment 8 - Task List, Milestones, Timeline, and Effort Matrix

Team Members: Nate Heath, Elliot Warner, William Thomas

Project Title: Data-Day

Goal Statement: To build and open source a 64 x 32 pixel desktop display and the developer platform for creating applications for the device.

Task List

1. Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module – Elliot Warner
2. Design the input-event interface for rotation and press in the Dial Button module – Elliot Warner
3. Develop dial rotation and button press detection when debouncing for the Dial Button module – Elliot Warner
4. Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module – Elliot Warner
5. Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module – Elliot Warner
6. Research operating-system and service requirements for application management on the Orange Pi module – William Thomas
7. Design the application lifecycle and state transitions for the Orange Pi module – William Thomas
8. Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module – William Thomas
9. Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module – William Thomas
10. Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module – William Thomas
11. Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module – Nate Heath
12. Design the framebuffer format for the Display module – Nate Heath
13. Develop framebuffer operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module – Nate Heath
14. Test edge coordinates, malformed frames, and refresh stability in the Display module – Nate Heath
15. Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device – Nate Heath

Milestones

1. [COMPLETE] Team formation and basic project description – 9/9/26

   Report the team's members and a simple project overview for leading idea that includes the problem statement and proposed solution with a general explanation of the implementation. Verified via Canvas submission by Professor Zhan and via email by project mentor Professor Aurisano.

2. [COMPLETE] High-level system design – 10/2/26

   Create a high-level overview of the system and how it will work without going into in-depth implementation details. Include team member responsibility for different parts of the system. Verified via Canvas submission by Professor Zhan.

3. [COMPLETE] Detailed system design complete – 10/14/26

   Build out lower-level implementation details from high-level system design, including pseudocode for important algorithms and specific tools to use. Include possible edge cases and responses to edge cases. Verified via Canvas submission by Professor Zhan.

4. Requirements list – 10/30/26

   List of all project requirements along with technical specifications for the entire project's scope. Requirements are specific, testable, verifiable, etc. Verified via Canvas submission by Professor Zhan.

5. Presentation – 11/18/26

   Overview of problem statement and stakeholders, followed by proposed solution at varying levels of depth, as well as future work and goals. Mentions potential roadblocks and proposed workarounds. Oral presentation accompanied by a slideshow. Verified in person and via Canvas submission by Professor Zhan.

6. Design notebook – 12/2/26

   Complete documentation of progress up until point of submission. Covers overall project goals, covers design decisions made throughout the project's lifecycle, team member roles, etc. Includes tracking documentation (timeline, GANTT chart, etc.) of future work. Verified via Canvas submission by Professor Zhan.

7. Working basic prototype (no SDK) – 1/7/27

The Orange Pi can display customizable pixels on the LED display in a comprehensible manner. Driver work is completed to a level of functionality where coded pixels can be displayed on the LED display. Verified via demo in team meeting by project mentor Professor Aurisano.

8. Define SDK methods – 2/2/27

   Comprehensive list of SDK methods/endpoints with descriptions of what they do. Does not include implementation. Pseudocode included where useful. Verified via discussion and demos in team meeting by all team members.

9. Complete SDK – 2/14/27

   Comprehensive list of SDK methods/endpoints are filled in with implementations with code. Code is commented and simple example snippets are provided when necessary. Verified via discussion and demos in team meeting by all team members.

10. A sample program can run on machine – 3/7/27

    A basic program using various SDK methods/endpoints displays an interactive program on the LED display correctly. Passes existing test suite. Verified via demo in team meeting by project mentor Professor Aurisano.

11. A user can upload a valid Python file – 3/15/27

    The local server functions as expected to allow a user to connect to the server and upload a Python file. The uploaded Python file passes through validation path, and invalid Python files are rejected/modified. Verified via demo in team meeting by project mentor Professor Aurisano.

12. Full path works – 3/22/27

    The entire path works as expected. The Python file uploads correctly onto the Orange Pi, which runs said file and displays the output on the LED display as expected. Verified via demo in team meeting by project mentor Professor Aurisano.

13. Demo programs complete – 3/29/27

    The demo programs using the SDK methods are completed and work as expected on the LED display. Demo programs pass test suites that ensure they don’t contain bugs that could surface during demos/presentations. Verified via demos in team meeting by project mentor Professor Aurisano.

14. Documentation complete – 4/5/27

Documentation of entire design process, from start to finish, is complete. Documentation is thorough and does not contain gaps in knowledge. Design decisions are recorded and diagrams included where necessary. Verified via discussion in team meeting by all team members.

15. Complete product – 4/15/27

    The product is completed, including the user documentation and exterior housing. An end-to-end demo of the system, including file upload, can be repeated regularly without issues. Verified via demo(s) in team meeting by project mentor Professor Aurisano.

Timeline

One row per Task List item with the task ID, statement, owner, start date, completion date, and dependencies, plus a row giving each milestone's planned completion date. Show your buffer explicitly: 15 to 20 percent per task plus a protected block before each major milestone. A Gantt figure is welcome in addition, but the table is the graded item.

Task ID	Statement	Owner	Start Date	Completion Date	Dependencies	Planned Completion Date	Buffer (Days)	Risk ID
T1	Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module	Elliot Warner	12/1/26	N/A	None	12/5/26	1	R5
T2	Design the input-event interface for rotation and press in the Dial Button module	Elliot Warner	12/5/26	N/A	None	12/10/26	1	
T3	Develop dial rotation and button press detection when debouncing for the Dial Button module	Elliot Warner	12/10/26	N/A	T1, T2	12/15/26	1	

T4	Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module	Elliot Warner	12/15/26	N/A	T3	12/20/26	1	
T5	Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module	Elliot Warner	1/4/27	N/A	T1, T2	2/9/27	7	
T6	Research operating-system and service requirements for application management on the Orange Pi module	Will Thomas	11/20/26	N/A	None	11/24/26	1	R4
T7	Design the application lifecycle and state transitions for the Orange Pi module	Will Thomas	12/5/26	N/A	T6	12/15/26	2	R4
T8	Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module	Will Thomas	2/8/27	N/A	T7	2/18/27	2	R4
T9	Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module	Will Thomas	2/18/27	N/A	T8	2/26/27	2	R4
T10	Develop the local upload and management	Will Thomas	2/26/27	N/A	T9	3/5/27	1	R4

	endpoints for installing and listing applications in the Orange Pi module							
T11	Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module	Nate Heath	11/25/26	N/A	T6, T7	11/30/26	1	R2,R5
T12	Design the framebuffer format for the Display module	Nate Heath	11/30/26	N/A	T11	12/7/26	1	R2
T13	Develop framebuffer operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module	Nate Heath	12/7/26	N/A	T12	12/16/26	2	R1,R2
T14	Test edge coordinates, malformed frames, and refresh stability in the Display module	Nate Heath	12/16/26	N/A	T13	12/20/26	1	R1,R2
T15	Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device	Nate Heath	3/5/27	N/A	T4, T14	3/15/27	2	R3

Note: Tasks may be blocked by tasks that have blocks so there may be more blocking areas than listed.

Effort Matrix

Task ID	Task	Primary Owner	Nate (hrs)	Elliot (hrs)	William (hrs)
T1	Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module	Elliot Warner	2	5	2
T2	Design the input-event interface for rotation and press in the Dial Button module	Elliot Warner	3	10	2
T3	Develop dial rotation and button press detection when debouncing for the Dial Button module	Elliot Warner	2	10	3
T4	Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module	Elliot Warner	2	12	3
T5	Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module	Elliot Warner	1	4	1
T6	Research operating-system and service requirements for application management on the Orange Pi module	William Thomas	2	2	10

T7	Design the application lifecycle and state transitions for the Orange Pi module	William Thomas	3	3	12
T8	Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module	William Thomas	5	5	26
T9	Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module	William Thomas	2	2	12
T10	Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module	William Thomas	3	3	18
T11	Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module	Nate Heath	10	3	2
T12	Design the framebuffer format for the Display module	Nate Heath	12	3	2
T13	Develop framebuffer operations for clearing, drawing pixels, rendering	Nate Heath	22	3	2

	text, and presenting frames in the Display module				
T14	Test edge coordinates, malformed frames, and refresh stability in the Display module	Nate Heath	12	3	2
T15	Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device	Nate Heath	14	3	2
TOTALS			95 hrs (34.5%)	71 hrs (26.8%)	99 hrs (38.7)

Work Breakdown Structure

Phase 1: Hardware and Low-Level Drivers

1.1) Input Hardware Investigation and Wiring (T1)

Task Statement: Investigate the dial/button hardware, GPIO pin requirements, and available input libraries for the Dial Button module.

Primary Owner: Elliot Warner

1.2) Display Driver Compatibility and Refresh Constraints (T11)

Task Statement: Investigate driver compatibility, wiring, power requirements, and refresh constraints for the Display module.

Primary Owner: Nate Heath

1.3) Framebuffer Format and Memory Allocation (T12)

Task Statement: Design the framebuffer format for the Display module.

Primary Owner: Nate Heath

1.4) Framebuffer Operations and Text Rendering (T13)

Task Statement: Develop framebuffer operations for clearing, drawing pixels, rendering text, and presenting frames in the Display module.

Primary Owner: Nate Heath

Phase 2: OS and Core Application Management

2.1) Operating System & Service Requirements (T6)

Task Statement: Research operating-system and service requirements for application management on the Orange Pi module.
Primary Owner: William Thomas

2.2) Application Lifecycle and State Transitions (T7)

Task Statement: Design the application lifecycle and state transitions for the Orange Pi module.
Primary Owner: William Thomas

2.3) Application Runtime Management (T8)

Task Statement: Develop application discovery, selection, launch, stop, and state saving handling for the Orange Pi module.
Primary Owner: William Thomas

2.4) Local Upload and Management Endpoints (T10)

Task Statement: Develop the local upload and management endpoints for installing and listing applications in the Orange Pi module.
Primary Owner: William Thomas

Phase 3: User Interface and Event Handling

3.1) Input-Event Interface Design (T2)

Task Statement: Design the input-event interface for rotation and press in the Dial Button module.
Primary Owner: Elliot Warner

3.2) Debouncing and Rotation Signal Handling (T3)

Task Statement: Develop dial rotation and button press detection when debouncing for the Dial Button module.
Primary Owner: Elliot Warner

Phase 4: Testing and Package Validation

4.1) Dial Input Boundary & Rapid Input Testing (T4)

Task Statement: Test boundary behavior, rapid input, and missed/duplicated events in the Dial Button module.
Primary Owner: Elliot Warner

4.2) Package Structure & Size Limit Validation (T9)

Task Statement: Validate package structure, metadata, file-size limits, and error responses in the Orange Pi module.
Primary Owner: William Thomas

4.3) Display Edge & Frame Stability Testing (T14)

Task Statement: Test edge coordinates, malformed frames, and refresh stability in the Display module.
Primary Owner: Nate Heath

Phase 5: Integration and Documentation

5.1) Module Documentation & Setup Guides (T5)

Task Statement: Document setup instructions, event details, and recovery behavior (if necessary) for the Dial Button module.
Primary Owner: Elliot Warner

5.2) Full System Integration & Device Testing (T15)

Task Statement: Integrate the Dial Button, Orange Pi, and Display modules and test application and selection on the complete device.
Primary Owner: Nate Heath

Risk Register

ID	Risk	Trigger	Mitigation	Owner	Likelihood	Impact	Strategy	Category	Taks
R1	Display work that slips past 12/20 can't be recovered over winter break	T13 not complete by 12/16	All three members work on the display build in parallel and share	Nate Heath	Medium	High	Reduce	Team	T13, T14

			findings so it's done before 12/20						
R2	Display driver/framebuffer work runs long	Any of T11–T14 misses its completion date	Demo a simpler version of the display output at M7 if the full framebuffer work isn't ready	Elliot Warner	Low	Medium	Accept	Technical	T11-T14
R3	Integration overruns its protected block	T15 not complete by 3/15	Cut a feature from the Full path works demo so the core path is shown on 3/22	Nate Heath	Low	Low	Accept	Technical	T15
R4	Critical path rests on one member	T6 not complete by 11/24	William cross-trains Nate and Elliot on the Orange Pi module by 1/14/27 so either can pick up T8–T10	Elliot Warner	Low	High	Reduce	Team	T6-T10

R 5	Hardware not on hand or incompatible	Hardware not operational by 12/1/26	All hardware is already on hand; if a part breaks, order a replacement immediately	Nate Heath	Low	High	Reduce	External	T1, T11